Электроника МКШ-2 — Display & Power Troubleshooting

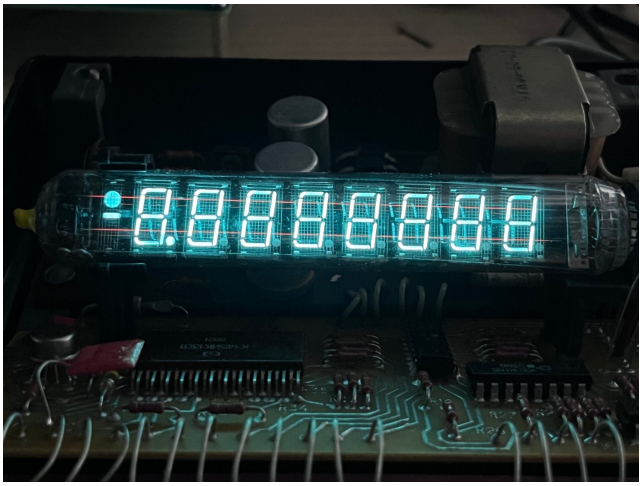
Workshop guide for the “all digits lit” fault and for setting up the 230V-to-42V converter. Part numbers and pin numbers come from the factory manual, appendix 1 (pages 29–30).

Unit / serial		Date	
Meter		Mains voltage at test	

Safety. Only connect or move probes and clip leads with the converter **unplugged**. Inside the calculator the highest DC rail is about 45–50V. Mains voltage is present only inside the converter, so keep its enclosure closed. Plug in through an RCD-protected outlet. Hot resistors and filament wiring can burn.

1. Symptoms and what they mean

- The IV-18 is **multiplexed**. Its 17 lines are 9 grids (one per position) and 8 segment lines shared by every digit. Each grid switches on in turn while the segments for that digit are driven.
- A line is switched **on** (to 0V) by a К161КН1Б driver and pulled back **off** to Г (−45V) by its own 56kΩ resistor (R1–R17). The filament sits a few volts above Г (set by R38/VD2), so a line at Г is cut off.
- A blank digit is made by driving **no** segments during its time slot. If lines are not pulled back to Г, the display shows stale data.
- If digit 1 shows a sensible “0.” and the display reacts to keys, the processor (К145ИК1301) and keyboard are working.



Pattern A: every digit shows the same “0.” at full brightness, with all decimal points lit. Red filament wires mean the supply is too high.



Pattern B: digit 1 correct; blank digits show copies of a real digit that fade away the further they are from it (here after typing 456).



Pattern C: after typing 3.1415, every position shows “9” and a minus appears in the sign position. “9” is all the segments of 3, 1, 4 and 5 combined: segments a–g are switched on but never switched off, while the decimal points behave correctly.

Display pattern	Most likely cause	Go to
C: every position shows the **combined segments** of all the digits entered (e.g. 3.1415 → “9” everywhere); a false minus appears once a digit uses the middle bar; decimal points correct	Segment lines a–g switch on but not off: weak or broken pull-down paths (bad solder joints), input or output side	Step 6 (6b, 6c, 6g), 6R
Correct digits for a moment after a keypress, then the other positions fill in over a few seconds	Driver inputs floating: input pull-downs (R19–R35, to −15V) not holding them off	Step 6 (6c, 6d, 6g), 6R
A: same character in every digit, full brightness	Grids not turning off: filament bias missing, or grid pull-downs not reaching Г	Steps 5, 6
B: blank digits show fading copies of a real digit	Segment lines not pulled back to Г: broken common return, or leakage	Step 6 (6b, 6c), 6R
Faint “ghost” segments behind real digits	Bias slightly low, supply too high, or board leakage	Steps 3, 5, 9
One segment or one digit always on or always off	One pull-down, one driver output, or one display lead	Step 6 (6a, 6b, 6c)
Pattern changes when the board is handled, or after reflowing	Dry joint, cracked track, or flux residue	Step 6 (6f), 6R
Visible red/orange filament glow	Supply too high	Steps 3, 8

2. Quick check: clear the calculator

☐ Switch on and press **Cx**. According to manual section 7.3, the calculator is ready when the display shows only “0.”.

☐ If anything else is displayed, switch off, wait **at least 3 seconds**, and switch on again (sections 7.3, 8.1.2).

☐ Repeat with a few entries and photograph the display each time. If digits that should be blank are lit, match the pattern to the table in step 1 and continue below.

Diagnostic entry (after Cx)	Correct display	If segments latch (pattern C)	Seen
(none)	“0.” in position 1, rest blank, no minus	“0” shapes in every position, no minus	
3 4	“34.”, rest blank	9-like shapes everywhere, false minus	
3 , 1 4 1 5	“3.1415”, rest blank	“9” in every position, false minus	
1 1 1	“111.”, rest blank	“1” in every position (no false minus)	

How to read it: the number builds from the left and the decimal point follows the last digit typed (manual Table 3). Positions to the right should be dark. If every position shows the same shape, work out which segments the entered digits use together: a match confirms pattern C.

3. Supply voltage at the calculator input

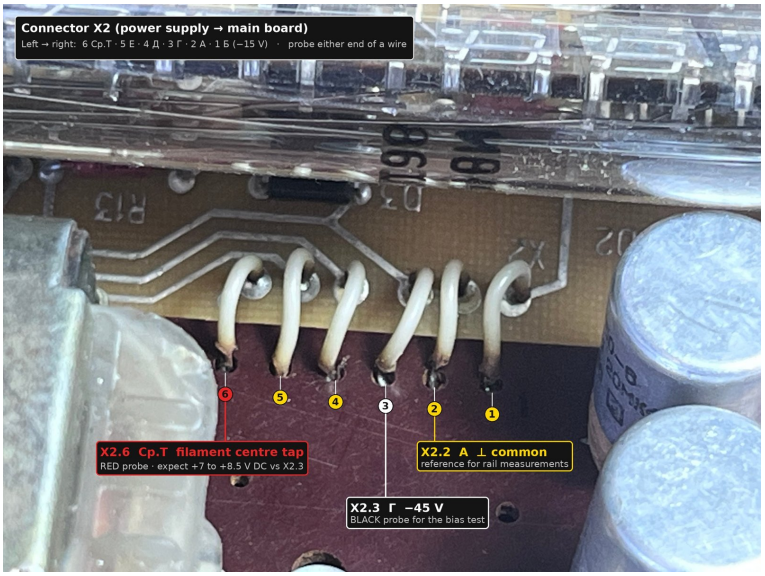
Specification (manual 3.10): **42V AC +10% / −15% at 50 ± 1Hz**, which allows **35.7–46.2V**. Power consumption is 9W at most (3.7), which is about 0.2A.

Measurement	Expected	Measured
Converter output, calculator connected and on (V AC)	42V (35.7–46.2V)	
Calculator input current (A AC, meter in series with one leg)	≤ 0.21A	

4. Internal supply rails at connector X2

Calculator running. Use the DC range unless marked AC, and measure relative to **X2 pin 2 (common, “+Общий”)** unless stated otherwise. The −15V rail is regulated. The −45V rail is **not** regulated and rises in proportion to the input voltage.

X2 pin	Net (schematic)	Expected	Measured
1	Б — −15V logic supply	≈ −15V DC (steady)	
2	А — common ⊥ (+Общий)	0V, reference	—
3	Г — −45V display drive	≈ −45V at 42V in; ≈ −50V at 48V in	
4 ↔ 5	Д / Е — IV-18 filament (Накал)	≈ 5V AC between pins 4 and 5	
6	Ср.Т — filament centre tap	See step 5	



Connector X2 on the main board, pins numbered from the right: 6 Ср.Т · 5 Е · 4 Д · 3 Г · 2 А · 1 Б. Each wire is one pin, so probe either end.

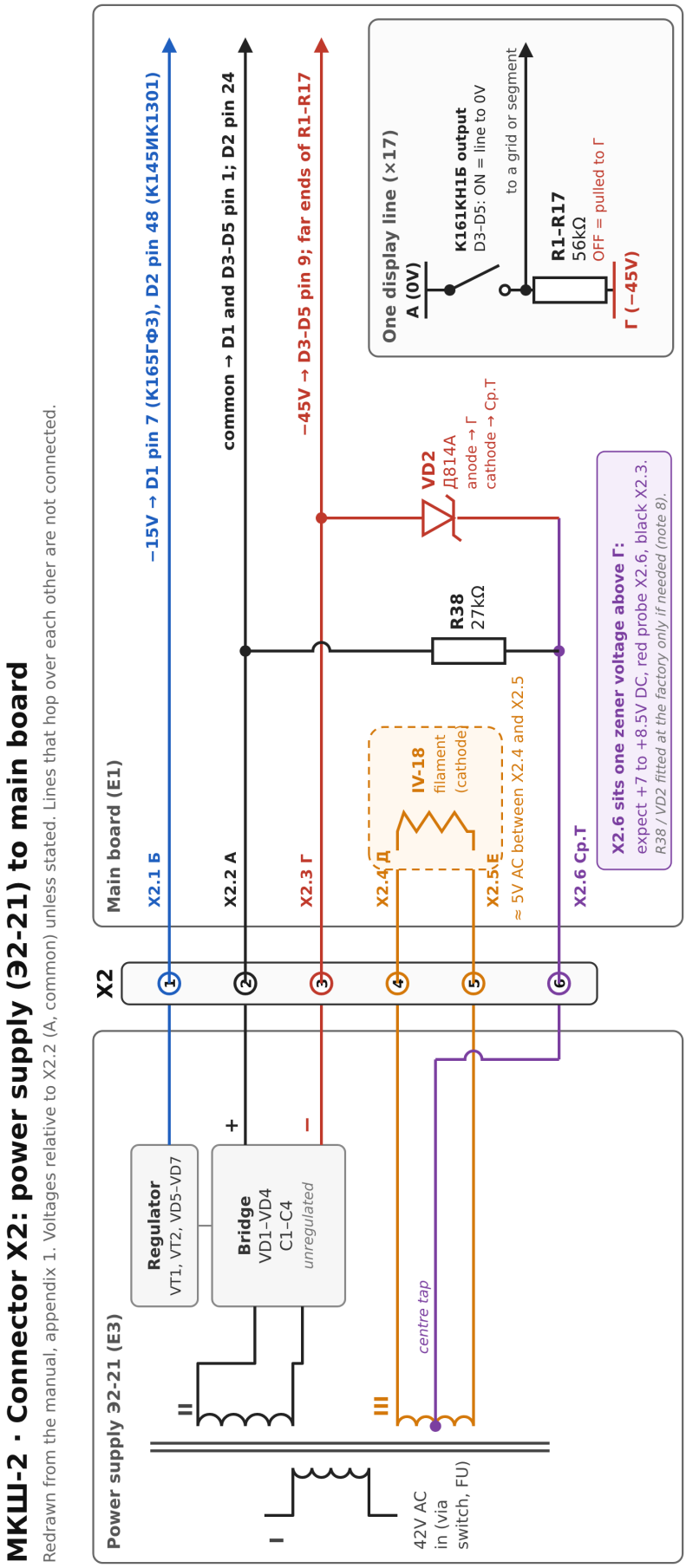
5. The key test: filament bias

The display’s 56kΩ pull-down resistors hold the “off” grids and segments at the −45V rail (Г). For them to stay dark, the filament must sit a few volts **above** −45V. R38 and VD2 set that level on the centre tap (X2 pin 6).

Measure (DC): black probe on **X2 pin 3 (−45V)**, red probe on **X2 pin 6 (centre tap)**.
Expect: about +7 to +8.5V (the Д814А zener voltage).
Measured: _____ V Polarity confirmed (+ on X2.6)? ☐ Yes ☐ No

Reading (pin 6 relative to pin 3)	Likely cause	Action
About +7 to +8.5V	Bias is correct	Go to step 6 (pull-downs and drivers)
About 0V, or negative	VD2 shorted, or centre tap tied to −45V	Replace VD2 (step 7)
Low or drifting	R38 open (zener gets no current)	Replace R38 (step 7)
R38 / VD2 not on the board	Not fitted at the factory (they were fitted only when needed)	Fit both (step 7)

The X2 interface, including R38 and VD2, is drawn in full on the next page (turn the page sideways).



Connector X2, redrawn from the manual (appendix 1, pages 29–30).

R38 (27kΩ) runs from X2.6 to common (A). VD2 (Д814А) has its **anode on Г (X2.3)** and its **cathode on X2.6**, so the filament centre tap sits one zener voltage above Г.

Note 8 of the schematic (translated): “Fitted during adjustment at the factory if unused digits of the display glow.”

The inset shows how each of the 17 display lines is switched on by a К161КН1Б output and pulled off to Г by its 56kΩ resistor.

6. Pull-downs, drivers and solder joints (if the bias is correct)

Each of the 17 display lines passes through **two pull-downs**, and both must work for a line to switch off:

- **Input pull-downs, R19–R35** (56kΩ, in a row between the processor D2 and the drivers). The processor switches a driver **on** by pulling its input to common (0V); the input pull-down to **Б (−15V)** is the only thing that switches it **off** again. If its path is broken, the driver input floats: the line switches on correctly, then is never properly switched off and drifts on over a few seconds.
- **Output pull-downs, R1–R17** (56kΩ, next to the display). The К161КН1Б output connects the line to common to light it; the output pull-down to **Г (−45V)** holds it below the filament when off. If its path is weak, the line switches off only slowly, or not at all.
- **Measure through the joints, not across the resistor.** A resistor can read 56kΩ across its leads while a cracked or dry joint at the pad leaves its path open or high. Tests 6b and 6c measure from the IC or display pin to the supply pin, so every joint in the path is included.

Driver mapping (traced on a working board). Output on pin p; its input is on pin 18 − p (manual: inputs 7 to 1 on pins 10–16). Confirm a couple of the input pins by following the track back to D2.

Line	IV-18 pin	Driver output	Driver input	Line	IV-18 pin	Driver output	Driver input
a	12	D5 pin 5	D5 pin 13	G9 (sign)	14	D5 pin 8	D5 pin 10
b	10	D5 pin 7	D5 pin 11	G8 (digit 1)	18	D3 pin 7	D3 pin 11
c	4	D5 pin 2	D5 pin 16	G7	19	D3 pin 6	D3 pin 12
d	3	D4 pin 4	D4 pin 14	G6	20	D3 pin 5	D3 pin 13
e	5	D4 pin 3	D4 pin 15	G5	17	D3 pin 4	D3 pin 14
f	11	D5 pin 6	D5 pin 12	G4	21	D3 pin 3	D3 pin 15
g	9 (and 7)	D4 pin 2	D4 pin 16	G3	16	D3 pin 2	D3 pin 16
dp	2	D4 pin 5	D4 pin 13	G2 / G1	22 / 15	D4 pins 7 / 6	D4 pins 11 / 12

Segments are split across D4 and D5, and both chips also drive lines that work, so a fault affecting a–g together is rarely one driver chip. Unused outputs: D3 pin 8, D4 pin 8, D5 pins 3 and 4. IV-18 pin 7 is tied to pin 9 on the board; pins 6 and 8 are not connected.

#	Test	Expected	Measured
6a	**Power off.** Resistance across each of R1–R17 and R19–R35, in circuit	≈ 53–58kΩ (a little under 56kΩ from parallel paths)	Lowest ____ Highest ____
6b	**Power off. Output path:** each IV-18 segment pin (12, 10, 4, 3, 5, 11, 9) to X2.3 (Г). Repeat for pin 2 (dp) as the reference	≈ 56kΩ each; same as dp	a__ b__ c__ d__ e__ f__ g__ dp__
6c	**Power off. Input path:** each driver input pin (table above) to X2.1 (−15V). Repeat for D4 pin 13 (dp) as the reference	≈ 56kΩ each; same as dp	a__ b__ c__ d__ e__ f__ g__ dp__
6d	**Power on, DC.** Shared end of the R19–R35 row relative to X2.2	≈ −15V (same as X2.1)	
6e	**Power on, DC.** Shared end of the R1–R17 row relative to X2.2	Same as X2.3 (≈ −45 to −49V)	
6f	**Flex / tap test.** Meter on 6b or 6c while flexing the board gently; then power on and tap around the resistor rows, D2–D5 and the IV-18 leads with an insulated stick	Steady reading; display does not change	Location: _____

#	Test	Expected	Measured
6g	**Drift test, power on, DC.** Watch a segment driver input (e.g. g, D4 pin 16) for a few seconds after a keypress; compare with dp (D4 pin 13)	Steady, like dp. A reading that creeps = floating input	☐ Steady ☐ Creeps
6h	**Stronger pull-down test.** Power off, tack 22kΩ across one segment's output pull-down; power on and type a digit that does not use that segment	Segment now switches off = its path was weak. Remove afterwards	Segment ____ ☐ Off ☐ No change
6i	**Power on.** К161КН1Б D3, D4, D5 pin 9 / pin 1 relative to X2.2	Pin 9 ≈ −45 to −49V; pin 1 = 0V	D3 ____ D4 ____ D5 ____
6j	**Scope (optional).** Segment and grid lines, compared with manual trace 5 (Д4 with "0" displayed)	Sharp pulses to 0V, straight back to ≈ −45V; no slow decay	

☐ 6b or 6c high, open or unsteady: **resolder that path** (step 6R below), then measure again.

☐ 6d or 6e not at its rail: find the break in that row's shared track back to X2.1 or X2.3; bridge it with a short wire if needed.

☐ 6g creeps: the input pull-down for that line is not holding it. Resolder that resistor and the driver input pin, then clean.

☐ All paths measure correctly but lines still fail to switch off: suspect leakage (clean thoroughly, step 6R) or a К161КН1Б output stuck on.

6R. Resoldering: often the actual fix

On this board, segments that would not switch off were fixed by resoldering the pull-down resistors. Joints this old often look sound but are cracked or dry underneath, and a quick touch-up is often not enough. The input and output pull-downs and the driver inputs are high-impedance nodes, so **flux residue left after reflowing can itself bring the fault back**.

☐ **Protect the MOS chips.** К145ИК1301 and К161КН1Б are static-sensitive: use a temperature-controlled, grounded iron and touch something grounded first.

☐ **Remove the old solder** with desoldering wick from each joint in the suspect path: both ends of the pull-down resistors, the driver pins, and the IV-18 lead joints.

☐ **Refill with fresh solder and flux,** heating each joint until the solder flows onto both the lead and the pad.

☐ **Check for bridges** between adjacent IC pins under a magnifier. Power off, adjacent driver input pins (10–16) should read several megohms or more apart.

☐ **Clean thoroughly:** scrub away all flux with isopropyl alcohol (99%) and a stiff brush, top and bottom of the board, and let it dry completely before powering up.

☐ **Re-test** with the diagnostic entries in step 2, and repeat 6b and 6c to confirm every path now measures about 56kΩ and stays steady when flexed.

7. Fit or replace R38 and VD2 (only if step 5 failed)

Ref	Original	Modern replacement	Notes
R38	27kΩ МЛТ-0.125	27kΩ 0.25W metal film	Value chosen at the factory (**).
VD2	Д814А zener, about 7–8.5V	8.2V 0.5–1W zener: 1N4738A or BZX79-C8V2	If faint glow remains, try 9.1V, then 10V.

☐ Unplug. Fit VD2 with its **anode (unbanded end) to X2.3 (Г)** and its **cathode (banded end) to X2.6 (Ср.Т)**. Fit R38 between X2.6 and common (A).

☐ Power up and repeat the step 5 measurement. Expect about the zener voltage.
Measured after fitting: _____ V

☐ Display shows a single “0.” and the other digits are dark.

☐ Don’t use more bias than you need: a higher zener also dims the segments that should be lit.

8. Series dropper resistor and output fuse (bring the supply down to 42V)

A converter that delivers more than 42V under load overdrives the IV-18 filament (visible red glow) and raises the −45V rail. A series resistor brings both back into specification. On its own it does not cure the display fault.

- **Series only.** Use one resistor in either output leg, or split the value across both legs (still in series). Never wire them in parallel: 22Ω ‖ 27Ω is only about 12Ω.
- Fit it **inside the converter**, not inside the calculator. Order on the converter output: **transformer → fuse → resistor → socket**.
- Choose the rating from **P = I² × R**, and use a resistor rated for at least 2× the calculated power, preferably 3× or more.
- **The calculator’s current depends on the display fault.** Lit segments draw current, so the current can fall noticeably once the display is fixed. Size the final dropper after the repair, measuring with the dropper fitted.

Method

☐ Calculator running, dropper fitted. Measure the AC voltage **across each resistor**: I = V ÷ R. Both should give the same current (this also confirms a series connection).

☐ Measure the transformer output **before** the dropper (V_T) and the voltage **at the calculator**, after the dropper.

☐ Required total: R = (V_T − 42) ÷ I. Refit, then measure again at the calculator, and check the filament (X2.4–X2.5 ≈ 5V AC).

Option (example: V_T ≈ 50V, I ≈ 81mA)	**Total**	**Power**	**Expected at calculator**	**Notes**
100Ω 3W wirewound or metal oxide	100Ω	≈ 0.65W	≈ 42–42.7V	Single part
2 × 51Ω 3W (e.g. TE/Neohm 5-2176793-0), one per leg	102Ω	≈ 0.33W each	≈ 42–42.7V	Same layout as a two-leg test rig
33Ω + 68Ω, each ≥ 1W	101Ω	≈ 0.22W + 0.45W	≈ 42.7V	From common values

Fuse	**Rating**	**Why**
Output	**About 2× the running current, slow-blow, 250V**, 5×20mm, same leg as the resistor, before it. Example: 160mA T with a ≈ 100Ω dropper	A short is limited by the dropper (≈ V_T ÷ R, ≈ 0.5A here). The fuse must be well under a third of that, so it blows in seconds before the resistor overheats.
Input (existing)	200mA slow-blow (230V); about 400mA slow-blow for a 115V build	Mains-side faults only. It cannot see an output short.

R = (V_T − 42) ÷ I **P** = I² × R Example: V_T = 50V, I = 0.081A → R ≈ 100Ω, P ≈ 0.65W

V_T = _____ V V across each resistor: _____ / _____ V

I = _____ A R = _____ Ω Fitted: _____ Ω / ____ W Fuse: _____ mA

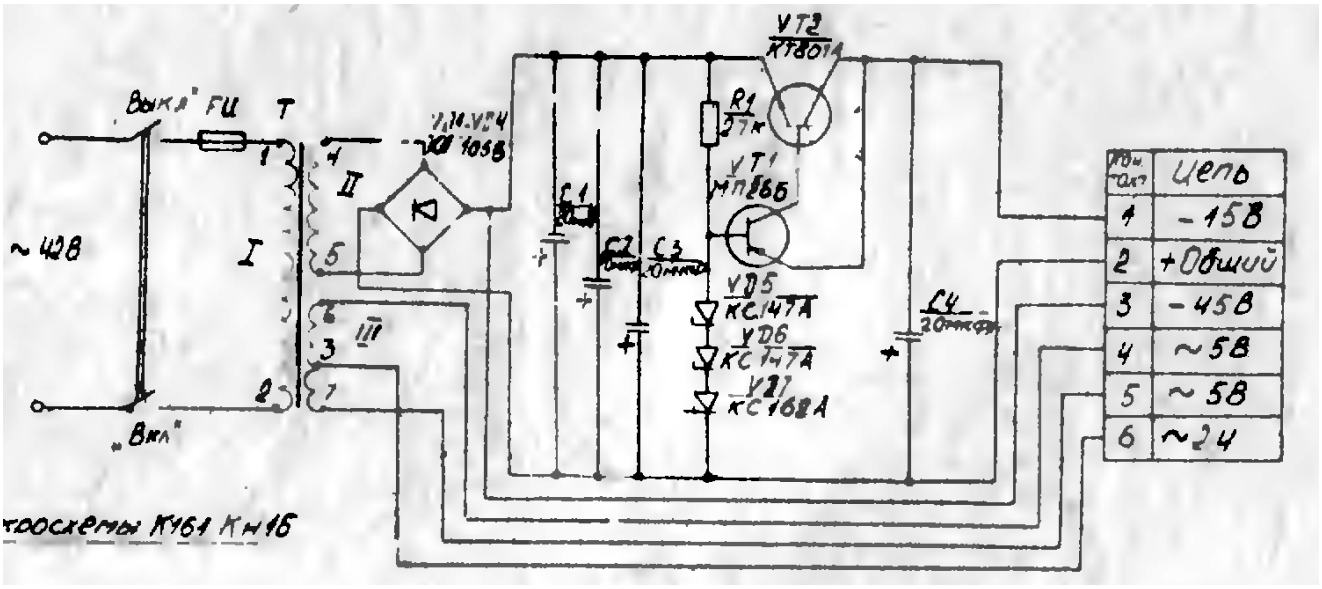
Voltage at calculator after fitting: _____ V Filament X2.4–X2.5: _____ V AC (target ≈ 5V)

Target: about 42V at the calculator with it running. The output still follows the mains, so ±1–2V through the day is normal and within specification. The filaments should no longer glow visibly red.

9. If the display is still wrong

- ☐ **Clean the main board.** Old flux, dust and moisture between the closely spaced −45V tracks can hold display lines partly on. Unplug, then scrub with isopropyl alcohol (99%) and a soft brush around D3–D5, R1–R17, the IV-18 leads and X2. Let it dry completely (at least an hour) before powering up. Note the display pattern before and after.
- ☐ **Electrolytics C1–C4** (K50-6, 100V) in the power supply block: check ESR, or measure AC ripple on the −15V and −45V rails. More than a few hundred mV is suspect. Ripple on −45V weakens the cutoff.
- ☐ **К161КН1Б drivers D3–D5:** with the step 6 tests passing, an output stuck on points to a failed driver.
- ☐ **Solder joints:** check the IV-18 wire leads, the driver chips and the X2 connector under a magnifier, and reflow anything doubtful. Also check the insulation of the X2 wires for heat damage or cracks where they are soldered.
- ☐ Final check: run the **74-step test** in manual Table 3 (pages 11–13) and compare the display after each key.

Reference: power supply block Э2-21 (manual page 30)



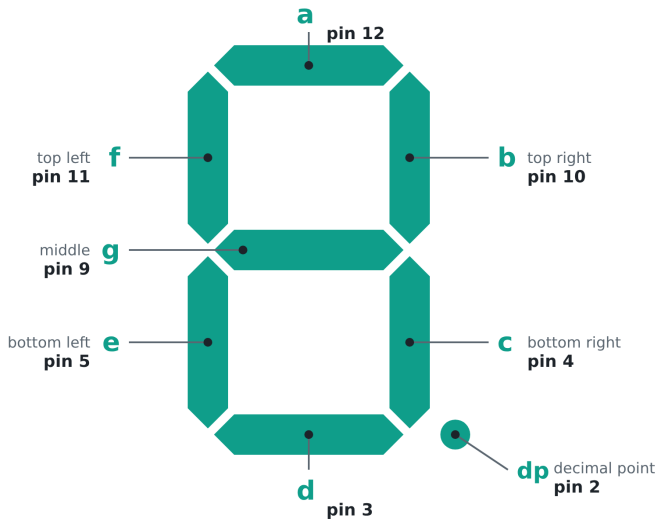
Ref	Part	Function
T, FU	Transformer, fuse	42V AC in. Winding II feeds the bridge; winding III is the filament winding with centre tap.
VD1–VD4	КД105В	Bridge rectifier, giving the unregulated −45V rail
C1–C4	К50-6, 100V	Filter and reservoir electrolytics. Suspect them first in a unit this old.
VT1, VT2	МП26Б, КТ801А	Series regulator for the −15V rail
R1	27kΩ	Regulator bias
VD5, VD6, VD7	КС147А, КС147А, КС168А	Reference chain for −15V (about 4.7 + 4.7 + 6.8V). The manual lists alternatives (Д814А, КС156А, etc.).

Item	Value
IV-18 filament	5V, 85mA typical (75–95mA), directly heated; pins 1 and 13
IV-18 currents	About 11mA per grid; about 8mA for all segments of one digit; 25–30mA with everything lit
Display drive	D2 output pulls the driver input to 0V (on); R19–R35 56kΩ to −15V switch it off. К161КН1Б output connects the line to 0V (on); R1–R17 56kΩ to −45V switch it off
Display lines	9 grids + 7 segments + decimal point = 17 lines = R1–R17 (pinout below)
Filament bias	X2.6 ≈ +7 to +8.5V relative to X2.3 (R38 27kΩ to common; VD2 Д814А, anode to X2.3, cathode to X2.6)
Logic	К145ИК1301 processor and К165ГФ3 clock generator on −15V
Supply input	42V AC +10% / −15%, 50 ± 1Hz, ≤ 9W

Reference: IV-18 (ИВ-18) pinout

IV-18 (ИВ-18) segments and pins

Segment letters follow the usual seven-segment convention. Pin numbers from the IV-18 data, cross-checked with the МКШ-2 manual.



Tube layout (front view, left to right)



G9	G8	G7	G6	G5	G4	G3	G2	G1
pin 14	pin 18	pin 19	pin 20	pin 17	pin 21	pin 16	pin 22	pin 15

Position G9 (far left): its dot is the dp line (pin 2) and its minus is segment g (pin 9). The МКШ-2 uses it for the sign.

G8 is the leftmost digit (digit 1); G1 is the rightmost.

Each of the 8 segment lines (a–g, dp) is shared by all positions; each grid (G1–G9) selects one position.

Filament (cathode): pins 1 and 13.

Segment	Pin	Position	Grid	Pin
a (top)	12	Sign / service (far left)	G9	14
b (top right)	10	Digit 1 (leftmost)	G8	18
c (bottom right)	4	Digit 2	G7	19
d (bottom)	3	Digit 3	G6	20
e (bottom left)	5	Digit 4	G5	17
f (top left)	11	Digit 5	G4	21
g (middle, also the minus)	9	Digit 6	G3	16
Decimal point	2	Digit 7	G2	22
Filament (cathode)	1, 13	Digit 8 (rightmost)	G1	15

Segment layout: a top, b top right, c bottom right, d bottom, e bottom left, f top left, g middle. Cross-check with the manual’s character table: “0” = pins 3, 4, 5, 10, 11, 12 (a–f); “1” = 4, 10 (b, c); “7” = 4, 10, 12 (a, b, c). The Russian sketch in some datasheets uses Cyrillic segment letters that do not map one-to-one onto a–g, so go by pin number.

Compiled from the Электроника МКШ-2 operating manual (Svetlovodsk, appendix 1), the IV-18 data compiled by Dieter Waechter (tube-tester.com, Reflector/Sovtek IV-18) and bench observations. The schematic scans are low quality, so check part values and orientations against the board before replacing anything.